

Abdelrahman O. Sawy

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EDUCATION

Rutgers University, New Brunswick, NJ

September 2025 – Present

Ph.D. in Industrial and Systems Engineering

Rutgers University, New Brunswick, NJ

September 2024 – January 2026

M.S. in Industrial and Systems Engineering · Concentration: System Analytics

GPA: 4.00 / 4.00

The American University in Cairo (AUC), New Cairo, Egypt

September 2019 – June 2024

B.S. in Mechanical Engineering · Concentrations: Design and Industrial Engineering

RESEARCH INTERESTS

Topics: Mathematical optimization, machine learning, operations research, spatio-temporal analytics, stochastic modeling, data-driven decision-making

Applications: Offshore wind energy operations & maintenance, renewable energy systems, predictive maintenance, supply chain optimization, AI-based forecasting

RESEARCH EXPERIENCE

Graduate Research Assistant

Spring 2025 – Present

Renewables & Industrial Analytics (RIA) Group, Rutgers University

Advisor: Prof. Ahmed Aziz Ezzat

- Conduct research on AI-based energy forecasting, offshore wind farm operations & maintenance (O&M) optimization, and spatio-temporal data analytics for renewable energy systems.
- Developed computational pipelines using the POSYDON stochastic maintenance scheduling framework to simulate multi-turbine offshore wind farm O&M strategies; validated simulation outputs against industry-standard platform Shoreline.io, producing statistically comparable performance benchmarks across key O&M metrics (availability, downtime, cost).
- Designed predictive maintenance scheduling models integrating turbine failure-rate distributions, weather-window constraints, and crew-vessel logistics to minimize unplanned downtime and lifecycle O&M costs.
- Applied spatio-temporal data science methods to analyze wind farm operational data, identifying patterns in failure modes, degradation trajectories, and seasonal maintenance demand across offshore assets.
- Contributing to ongoing research targeting journal-quality manuscripts on AI-driven O&M optimization and data-driven maintenance decision-making for offshore wind energy.

PUBLICATIONS & SCHOLARLY OUTPUT

Manuscripts in Preparation

- [P1] A. Sawy, K. Wang, and A. A. Ezzat, “Deterministic and Stochastic Optimization Approaches for Offshore Wind Farm Maintenance and Production Planning,” *manuscript in preparation*, 2026.

Note: Active contributor to the RIA Research Group’s publication pipeline on offshore wind O&M optimization and spatio-temporal energy analytics.

PRESENTATIONS & POSTERS

- [1] A. Sawy, K. Wang, and A. A. Ezzat, “Deterministic and Stochastic Optimization Approaches for Offshore Wind Farm Maintenance and Production Planning,” *ISE Research Day*, Rutgers University, New Brunswick, NJ, 2026. (Poster)

TEACHING EXPERIENCE

Graduate Teaching Assistant

January 2026 – Present

Computational Methods in Industrial Engineering (Graduate Level)

Rutgers University, Department of Industrial & Systems Engineering

- Deliver instruction on multivariate statistics, principal component analysis (PCA), regression methods, missing data imputation, recommendation systems, graph mining, Monte Carlo integration, and Markov Chain Monte Carlo (MCMC) to 30+ graduate students.
- Develop structured instructional materials including analytical derivations (eigenvalue decomposition, PCA optimization formulation), Python/Excel-based computational templates, and machine learning implementation examples.
- Provide one-on-one academic support and hold office hours to reinforce computational methods and data analytics concepts.

Graduate Teaching Assistant

September 2025 – January 2026

Probabilistic Models in Operations Research (Undergraduate Level)

Rutgers University, Department of Industrial & Systems Engineering

- Supported instruction in decision-theoretic models, Markov chain analysis, queuing theory, and systems under uncertainty for undergraduate engineering students.
- Developed Excel-based instructional materials for stochastic decision models and guided students through implementation of Markov/queuing systems.

HONORS & AWARDS

- **Perfect GPA (4.00/4.00)**, M.S. in Industrial and Systems Engineering, Rutgers University, 2026.
- **Kaggle Machine Learning Competition — Top 0.6%**, Ranked 21 out of 3,724 participants in loan default prediction competition, 2025.

PROJECTS & COMPETITIONS

Loan Default Prediction — Kaggle ML Competition

2025

- Ranked **21 out of 3,724 participants (top 0.6%)**; designed and tuned XGBoost and stacked ensemble models with advanced feature engineering to predict loan repayment outcomes on large-scale financial datasets.

Electricity Price Forecasting — P&G / IISE Forecasting Competition

2025

- Built electricity spot-price forecasting model using Locational Marginal Pricing (LMP) data; applied time-series methods (ARIMA, gradient boosting) with lag-feature engineering and cross-validation.

Depression Risk Classification — Kaggle ML Competition

2024

- Developed end-to-end ML pipeline with clinical feature engineering, missing data imputation, and ensemble classifiers (Random Forest, XGBoost) for mental health risk prediction.

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, R, SQL, LaTeX

Optimization & Operations Research: Gurobi, CPLEX, SciPy (LP/MILP/NLP), POSYDON framework, STOCHOS framework, mathematical programming, stochastic & deterministic modeling

Machine Learning & Data Science: scikit-learn, XGBoost, LightGBM, pandas, NumPy, Matplotlib, Seaborn, time-series forecasting, PCA, regression, MCMC, classification, clustering, missing data imputation

Simulation & Engineering Tools: ANSYS Fluent/Workbench, SolidWorks, CATIA, Monte Carlo simulation

Software & Platforms: Git/GitHub, Jupyter, RStudio, Microsoft Office Suite, Claude Code, LLM-based workflow automation

INDUSTRY EXPERIENCE

Engineer Intern — MDA Service Sector, B.Tech

Winter 2023

Cairo, Egypt

- Analyzed performance datasets for major domestic appliances (MDAs), producing diagnostic reports that identified system inefficiencies and informed data-driven maintenance strategies.
- Implemented predictive maintenance workflows in collaboration with service technicians, reducing unplanned service interventions and improving service throughput.

Engineer Intern — MTI Automotive Co.

Summer 2023

New Cairo, Egypt

- Conducted time-and-motion studies for engine, chassis, and bodywork repair operations; designed standardized operating procedures (SOPs) that improved service consistency and reduced rework.
- Optimized parts supply-chain operations by analyzing inventory usage trends, forecasting demand, and implementing just-in-time (JIT) principles for hybrid-vehicle battery management.

LEADERSHIP & ENTREPRENEURSHIP

Founder — The Agentic Lab®

2025 – Present

theagenticlab.net

- Founded AI automation startup delivering agentic workflow solutions and open-source tools.
Published at: github.com/AOSawy.
- Creating educational content on AI and automation trends across social media platforms, covering developments in artificial intelligence, agentic systems, and applied machine learning.

Marketing Head — AUC Developers Association

2022

- Led marketing strategy and outreach campaigns for the university's developer community; grew membership engagement and event participation.

Additional Activities: Member, AUC Mechanical Engineering Association — Racing & FabLab Teams (2024) · Student Recruiter, AUC CIMUN Model United Nations (2022) · Track & Field Team (2024)

SELECTED COURSEWORK

Optimization & Operations Research: Deterministic Models in Industrial Engineering · Engineering Decision Making Under Uncertainty · Supply Chain Engineering · Planning & Operations Engineering · Production Analysis

Data Science & Analytics: Computational Methods in Industrial Engineering · Data Analytics in Engineering Systems · Data Mining · Forecasting Analytics · Quality Management

REFERENCES

Prof. Ahmed Aziz Ezzat

Assistant Professor, Department of Industrial & Systems Engineering, Rutgers University

Faculty Affiliate, Rutgers Energy & Climate Institute (RCEI)

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Additional references available upon request.